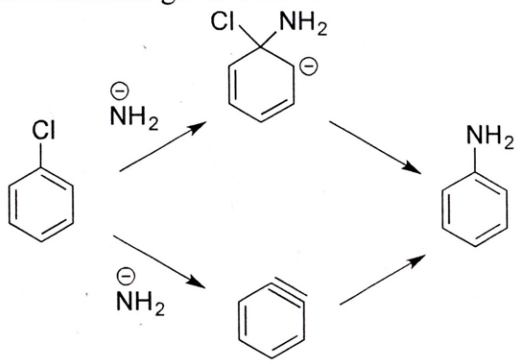
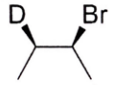
Organic Cume 03-31-2017

1. The reaction of benzoyl chloride with ethanol and pyridine to form ethyl benzoate and pyridinium chloride can proceed either by nucleophilic attack of the ethanol or pyridine in the first step. With nucleophilic attack by pyridine, an acyl-pyridinium species forms, so this kind of reaction amounts to nucleophilic catalysis by this added base. If ethanol attacks first, the pyridine only acts to scavenge the HCl formed in the reaction. Write these two possible mechanisms, and derive rate laws that distinguish these possibilities.

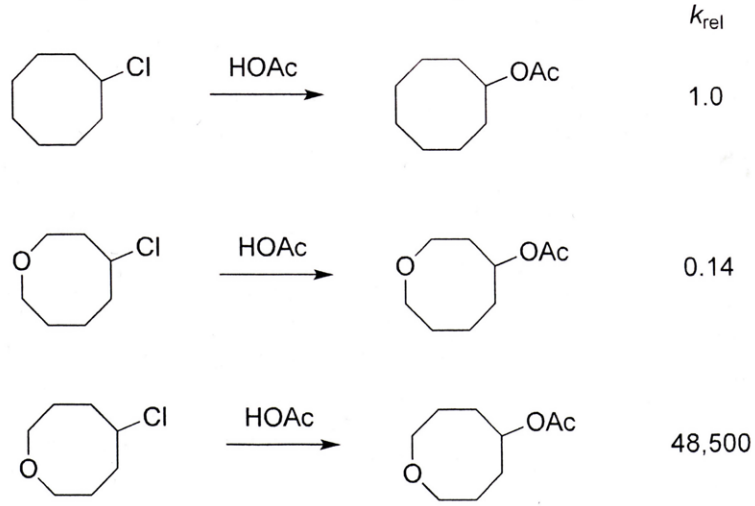
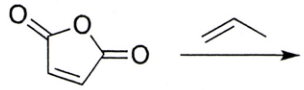
2. The treatment of chlorobenzene with potassium amide in liquid ammonia results in the formation of aniline. Propose four experiments discussed in this chapter (chapter 8 of Anslyn's book), or others you can devise, that can be used to distinguish between the two mechanisms given below.



3. The following alkyl bromide gives both *cis*- and *trans*-2-butene upon reaction with sodium ethoxide. Only one of these alkenes retains the deuterium label. Draw both products and explain why only one is deuterated.



4. Explain why placement of an oxygen in the chlorocyclooctane ring slows down the solvolysis reaction in one case and speeds it up in the other.



5. Predict the thermally allowed products of the following two reactions.